

PlotTwist

A Creative Plot Generation Framework with
Small Language Models

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I · THE PLEDGE II · THE TURN III · THE PRESTIGE



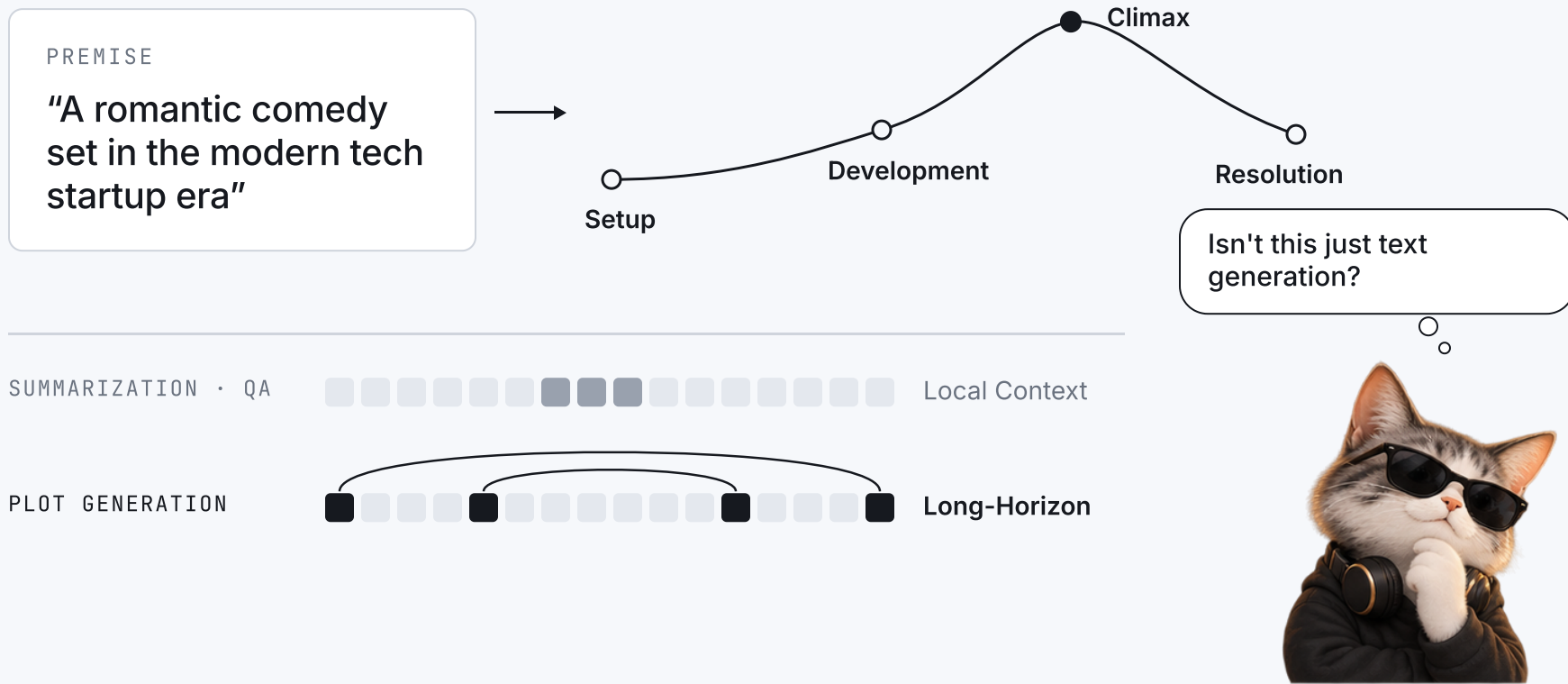
ACT I

The Pledge

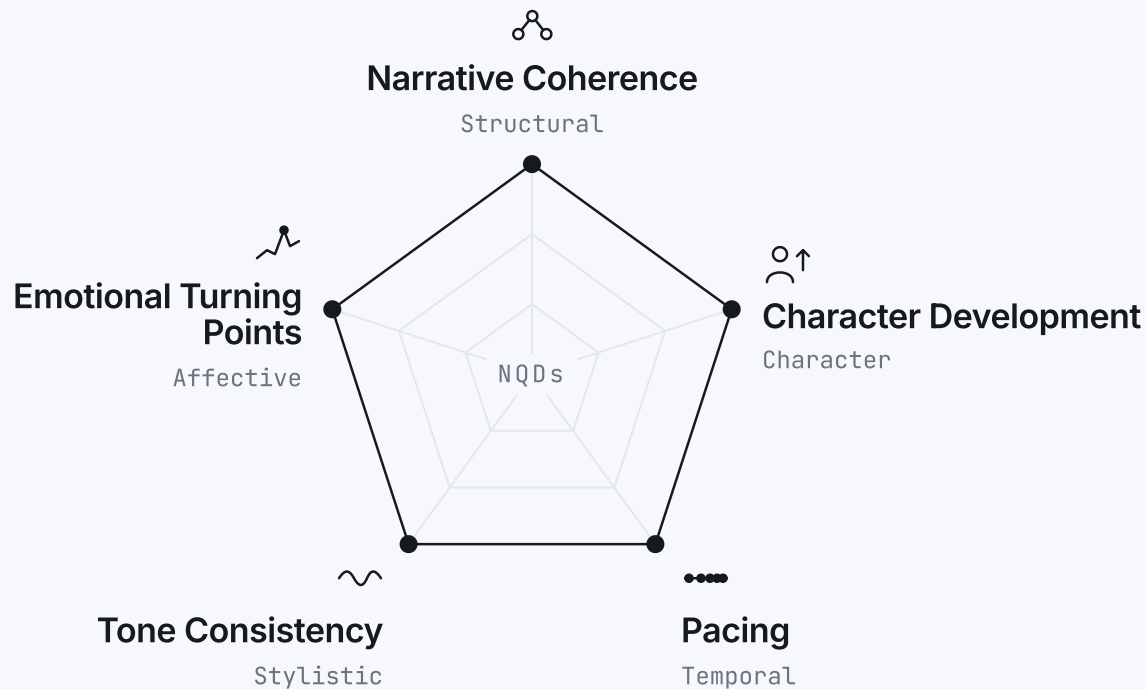
The task, and the standard it is held to.



A premise must expand into a full narrative arc



Five dimensions define plot quality



How do you score a story?



ACT II

The Turn

Why the task is hard, and how PlotTwist restructures it.



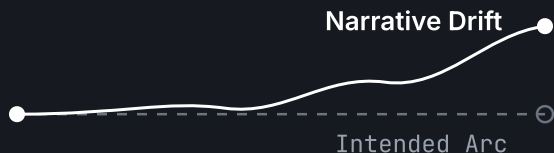
Aligning at scale is costly; LLM judges are biased

01 · GENERATION



Frontier Alignment

Computationally Prohibitive



Scale $\neq$ Long-Horizon Coherence

02 · EVALUATION



LLM-as-a-Judge

Systematic Biases



Positivity Bias



Same-Family Preference



Verbosity Bias



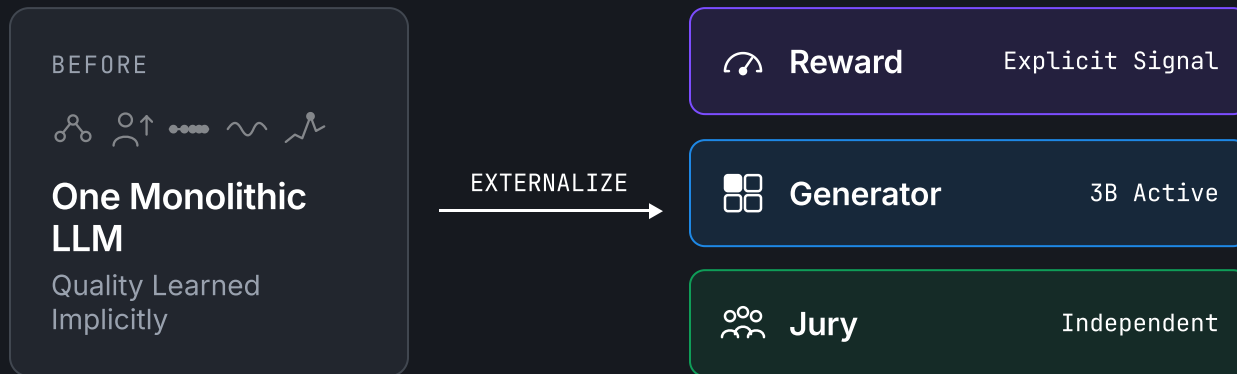
Limited Expert Agreement

+ Correlated Errors Across Models

Why not just use a bigger model?



PlotTwist externalizes structure instead of scaling



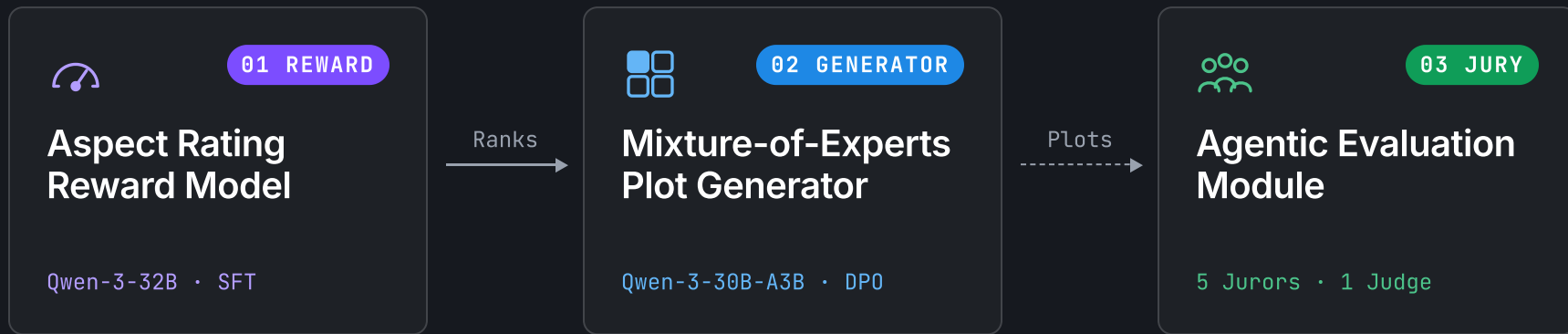
RESEARCH QUESTION

Can $\leq 3B$ -active SLMs match frontier LLMs?



Three modules divide the work

TRAINING



INFERENCE

Premise

→

Generator

→

Plot

1 Model · 1 Pass

Paired prompts counter rater positivity bias

01 REWARD



Why ask for the positives and negatives separately?

5,000 MovieLens Plots

→

5 NQDs × 2 Prompts × 5 Raters

→

SFT Qwen-3-32B · CE + Huber



DPO aligns a 3B-active generator on 160 pairs

02 GENERATOR

5,000 Premises

Base MoE + 3 Frontier LLMs Write Plots

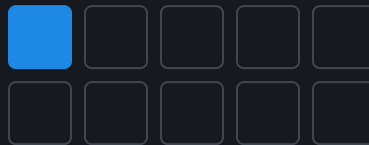
• Reward Model Scores Each Plot

Keep: Frontier Best · $> 8 \cdot \Delta \geq 0.5$

160 Pairs

DPO
→

Qwen-3-30B-A3B



3B / 30B Active

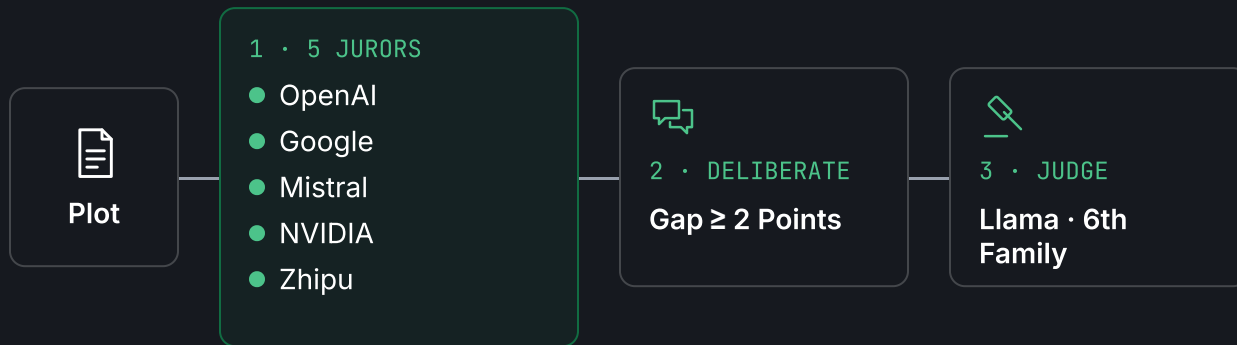
DPO Only · No SFT

Only 160 pairs?



A cross-family jury evaluates plots independently

03 JURY



Who judges the judges?



10 Criteria / NQD

Each 0-1 · Summed to 10



94-99% Verified

Evidence Quoted Verbatim



ACT III

The Prestige

What a model with 3B active parameters achieves.



Both evaluators separate classics from Razzies

101 Greatest Screenplays (GSAT) vs Razzie Winners · 1,000 Balanced Runs



Can they tell a classic from a Razzie?

✓ GSAT Higher in 100% of Runs · All 5 NQDs



The jurors agree on rankings more than on scores

SCHEMATIC · SAME ORDER, DIFFERENT LENIENCY



SCORE AGREEMENT

0.41 → 0.64

Krippendorff's α · Recentered

RANK AGREEMENT

0.98

Kendall's W



≈ 1.3 Effective Judges of 5

Correlated Judgments

✓ No Same-Family Cells · Same Ranking

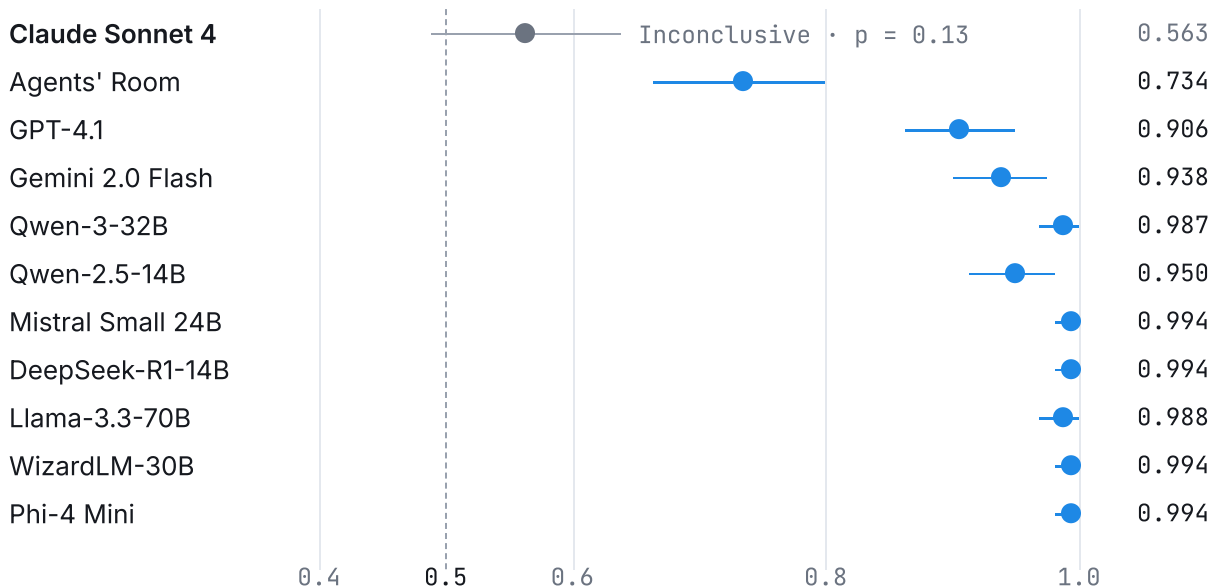
✓ 113 Specifications · Same Ranking

Are five judges really five?



PlotTwist wins against ten of eleven baselines

Paired Win Rate · 160 Premises · 95% CI · Ties = 0.5



MEAN JURY SCORE

PlotTwist **8.36**

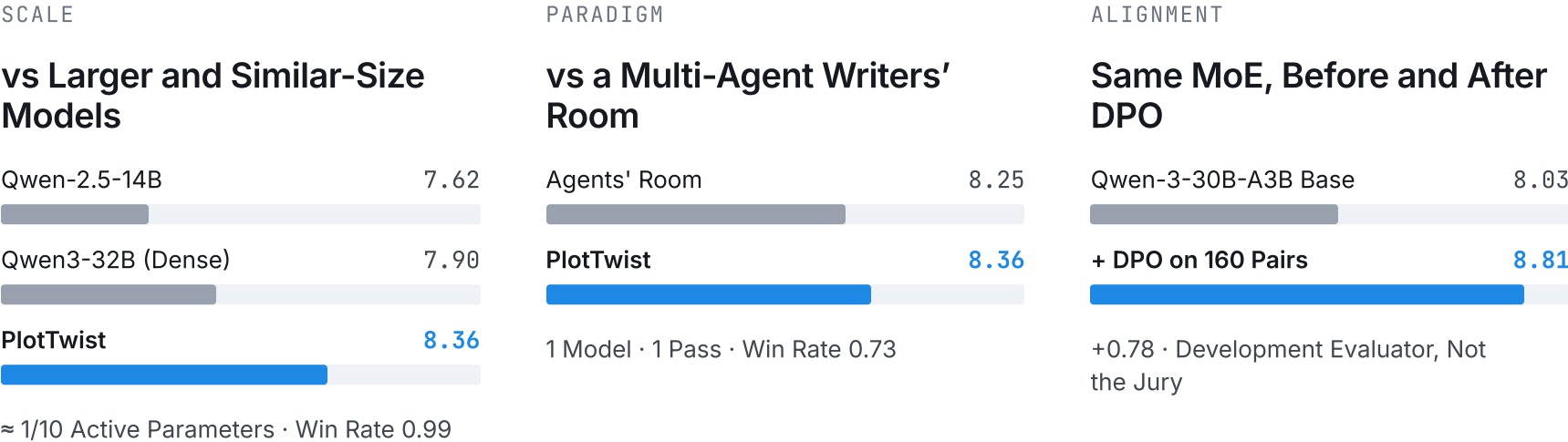
Claude Sonnet 4 **8.35**

Near-Tie in Mean → Win Rate
Is Primary



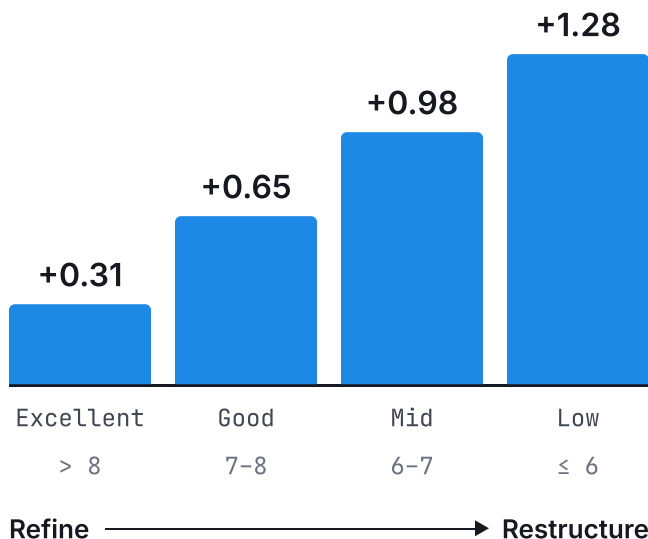
Neither scale nor paradigm explains the gains

Overall Mean Score · Axis 7–9



Gains grow as source quality drops

Gain over Source Plot · 40 Films per IMDb Stratum



EXCELLENT

Light Touch · Pacing Unchanged

LOW

Beats Original on $\geq 97.5\%$ of Films

Doesn't it just inflate every score?



3B active parameters, competitive with the frontier

Structured, preference-based alignment is a resource-efficient approach to high-quality creative plot generation.



01 REWARD

Positive–Negative Aspect
Ratings



02 GENERATOR

3B-Active MoE · DPO ·
160 Pairs



03 JURY

Cross-Family, Evidence-
Grounded

LIMITS

ⓘ LLM Judges, Not Experts

ⓘ English Synopses

ⓘ Claude Comparison Open



CURTAIN CALL

Any questions?



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Visit here to know
more about our work

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